

基于 YOLO11 识别红绿灯

三. 模型部署、验证及应用

4、模型部署及验证

远程连接到小车，在小车 userdata 目录下，新建 yolo11 文件夹。

把 best_bayese_640x640_nv12_modified.bin、classes.txt 上传到该目录下，运行（这是一条命令）

```
cp /opt/tros/humble/lib/dnn_node_example/config/yolov10workconfig.json
/userdata/yolo11/yolov11workconfig.json
```

打开 yolov11workconfig.json，修改

```
yolov11workconfig.j...
1{
2  "model_file": "/userdata/yolo11/best_bayese_640x640_nv12_modified.bin",
3  "task_num": 4,
4  "dnn_parser": "yolov11",
5  "model_output_count": 6,
6  "reg_max": 16,
7  "class_num": 3,
8  "cls_names_list": "/userdata/yolo11/classes.txt",
9  "strides": [8, 16, 32],
10 "score_threshold": 0.25,
11 "top_k": 300,
12 "output_order": [0,1,2,3,4,5]
13}
14
```

上传几张红绿灯图片到 yolo11 目录下，运行

export CAM_TYPE=fb

```
ros2 launch dnn_node_example dnn_node_example.launch.py
```

```
dnn_example_config_file:=/userdata/yolo11/yolov11workconfig.json
```

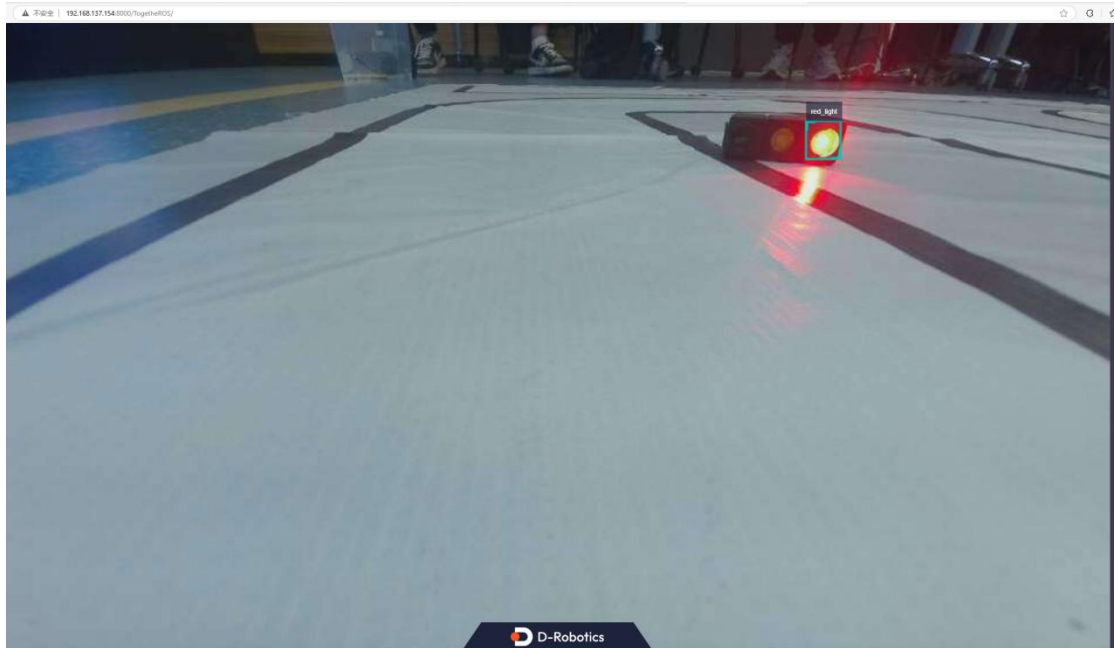
```
publish_image_source:=/userdata/yolo11/照片名
```

***黄色的是一条指令**

```
root@ubuntu:~# export CAM_TYPE=fb
root@ubuntu:~# ros2 launch dnn_node_example dnn_node_example.launch.py dnn_example_config_file:=/userdata/yolo11/yolov11workconfig.json pub
/target.jpg

[INFO] [launch]: All log files can be found below /root/.ros/log/2025-04-13-14-25-21-789200-ubuntu-4269
[INFO] [launch]: Default logging verbosity is set to INFO
dnn_node_example_path is /opt/tros/humble/lib/dnn_node_example
cp_cmd is cp -r /opt/tros/humble/lib/dnn_node_example/config .
camera_type is fb
using feedback
Hobot shm pkg enables zero-copy with fastrtps profiles file: /opt/tros/humble/lib/hobot_shm/config/shm_fastdds.xml
Hobot shm pkg sets RMW_FASTRTS_USE_QOS_FROM_XML: 1
env of RMW_FASTRTS_USE_QOS_FROM_XML is 1, ignore env setting
env of RMW_FASTRTS_USE_QOS_FROM_XML is 1, ignore env setting
launch webserver
pwd_path is /root
webserver_path is /opt/tros/humble/lib/websocket/websocket
launch webserver cmd is ./sbin/nginx -p .
env of RMW_FASTRTS_USE_QOS_FROM_XML is 1, ignore env setting
env of RMW_FASTRTS_USE_QOS_FROM_XML is 1, ignore env setting
env of RMW_FASTRTS_USE_QOS_FROM_XML is 1, ignore env setting
env of RMW_FASTRTS_USE_QOS_FROM_XML is 1, ignore env setting
env of RMW_FASTRTS_USE_QOS_FROM_XML is 1, ignore env setting
env of RMW_FASTRTS_USE_QOS_FROM_XML is 1, ignore env setting
```

在浏览器，输入小车 IP:8000 观察



中断程序后，输入

`export CAM_TYPE=usb`

`ros2 launch dnn_node_example dnn_node_example.launch.py`

`dnn_example_config_file:=/userdata/yolo11/yolov11workconfig.json`

***黄色的是一条指令**

```
root@ubuntu:~# export CAM_TYPE=mipi
root@ubuntu:~# ros2 launch dnn_node_example dnn_node_example.launch.py dnn_example_config_file:=/userdata/yolo11/yolov11workconfig.json
[INFO] [launch]: All log files can be found below /root/.ros/log/2025-04-13-14-31-26-855377-ubuntu-11149
[INFO] [launch]: Default logging verbosity is set to INFO
dnn_node_example path is /opt/tros/humble/lib/dnn_node_example
cp_cmd is cp -r /opt/tros/humble/lib/dnn_node_example/config .
camera_type is mipi
using mipi cam
Hobot shm pkg enables zero-copy with fastrtps profiles file: /opt/tros/humble/lib/hobot_shm/config/shm_fastdds.xml
Hobot shm pkg sets RMW_FASTRTPS_USE_QOS_FROM_XML: 1
config file path is /opt/tros/humble/lib/mipi_cam/config/
env of RMW_FASTRTPS_USE_QOS_FROM_XML is 1, ignore env setting
env of RMW_FASTRTPS_USE_QOS_FROM_XML is 1, ignore env setting
webserver has launch
env of RMW_FASTRTPS_USE_QOS_FROM_XML is 1, ignore env setting
config file path is /opt/tros/humble/lib/mipi_cam/config/
env of RMW_FASTRTPS_USE_QOS_FROM_XML is 1, ignore env setting
env of RMW_FASTRTPS_USE_QOS_FROM_XML is 1, ignore env setting
env of RMW_FASTRTPS_USE_QOS_FROM_XML is 1, ignore env setting
env of RMW_FASTRTPS_USE_QOS_FROM_XML is 1, ignore env setting
webserver has launch
```

把红绿灯放在小车前面，从网页观察



开另一个窗口，输入

`ros2 topic echo /hobot_dnn_detection`

```

time_ms_duration: 12.0
targets:
- type: red_light
  track_id: 0
  rois:
  - type: red_light
    rect:
      x_offset: 37
      y_offset: 34
      height: 91
      width: 84
      do_rectify: false
      confidence: 0.5831394791603088
    attributes: []
    points: []
    captures: []
- type: red_light
  track_id: 0
  rois:
  - type: red_light
    rect:
      x_offset: 70
      y_offset: 45
      height: 66
      width: 49
      do_rectify: false
      confidence: 0.2695736289024353
    attributes: []
    points: []
    captures: []
disappeared_targets: []

```

5、实际使用

1) 在 /userdata/dev_ws/install/lanedetection/share/lanedetection/目录下，新建 launch 文件夹，上传 dnn_node_yolo.launch.py 到这个目录下

2) 第 1 个窗口，启动底盘、摄像头和激光雷达

```
ros2 launch originbot bringup originbot.launch.py use_camera:=true use_lidar:=true
```

3) 第 2 个窗口，启动 yolo 识别红绿灯的程序

```
ros2 launch lanedetection dnn_node_yolo.launch.py
```

4) 第 3 个窗口，查看话题（topic）和数据

```
ros2 topic list
```

```
root@ubuntu:~# ros2 topic list
/camera_info
/cmd_vel
/hbmem_img0x2107020403080120021609080306270d
/odom
/originbot_status
/parameter_events
/rosout
/scan
/tf
/tf_static
/traffic_sign
/ydlidar_ros2_driver_node/transition_event
root@ubuntu:~#
```

查看数据

```
ros2 topic echo /traffic_sign
```

```
targets:
- type: "red_light\r"
  track_id: 0
  rois:
  - type: "red_light\r"
    rect:
      x_offset: 612
      y_offset: 55
      height: 41
      width: 41
      do_rectify: false
      confidence: 0.41854217648506165
    attributes: []
    points: []
    captures: []
  disappeared_targets: []
```

红灯 red_light

```
targets:
- type: green_light
  track_id: 0
  rois:
  - type: green_light
    rect:
      x_offset: 652
      y_offset: 67
      height: 32
      width: 31
      do_rectify: false
    confidence: 0.7247857451438904
  attributes: []
  points: []
  captures: []
disappeared_targets: []
```

绿灯 green_light

查看数据类型

`ros2 topic type /traffic_sign`

```
^Croot@ubuntu:~# ros2 topic type /traffic_sign
ai_msgs/msg/PerceptionTargets
root@ubuntu:~#
```

```
root@ubuntu:/userdata# python view_traffic.py
[INFO] [1744528531.407160094] [View_Traffic]: Traffic_sign is OK
green_light
green_light
green_light
green_light
green_light
```